Julien Fayolle

End-of-Studies Internship (6 months) — AI Engineer / LLMOps

Available from April 2027 · Open to International (East Asia focus) & France

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Professional & R&D Experience

Laboratoire CRIStAL (CNRS) & CHU de Lille · Lille, France *Since Sep. 2026*
AI R&D Project – Clinical RAG & Unstructured Data Processing

- Architecture and design of an on-premise clinical RAG pipeline leveraging the **MIMIC-IV** clinical database.
- Extraction and structuring of complex tabular and textual data using **Docling**; dense vectorization with **BGE-M3**.

BrioHR · Kuala Lumpur, Malaysia *Mar. – Aug. 2026 (6 mos.)*
AI Operations & Engineering Intern

- **Customer Support RAG Architecture:** End-to-end design of a RAG assistant connected to a cloud vector database; developed a custom web widget (JS, PHP) with HTTP session persistence.
- **Middleware Orchestration & Hybrid Routing:** Pipeline engineered in **n8n** with an intermediate LLM classifier to route tickets (60 % handled by AI, 40 % escalated to agents via webhooks and CRM API).
- **Security & Evaluation:** Secured API traffic via HTTP header signatures; evaluated real-world answer accuracy through a *Shadowing* deployment (Human-in-the-loop evaluation).

CHU de Lille · Lille, France *Jan. – Feb. 2026 (7 wks.)*
AI Engineer – Medical Computer Vision (Centrale Digital Lab Project, team of 3)

- Trained Vision Transformer models (**SwinUNETR**) in **PyTorch** with a custom MRI preprocessing pipeline for 3D ischemic stroke lesion segmentation in neuroradiology.
- Rigorous performance benchmarking (**Dice Score: 0.76** on ISLES2022 benchmark and **0.61** on real-world clinical datasets).

Cofidis · Lille, France *Nov. – Dec. 2025 (7 wks.)*
AI Engineer – MLOps & Application Deployment (Centrale Digital Lab Project, team of 4)

- Developed, containerized end-to-end with **Docker**, and delivered a turnkey carbon footprint RAG application; database interfacing and production-ready web integration.

Enedis · Lille, France *Sept. – Oct. 2025 (7 wks.)*
AI Engineer – NLP & Semantic Search (Centrale Digital Lab Project, team of 3)

- Designed and implemented a complete RAG pipeline *from scratch* in pure Python (without high-level abstraction frameworks): document chunking, embedding generation, and vector search engine for internal HR.

Education

École Centrale de Lille · Lille, France *Since Sep. 2023*
Master of Science in Engineering (Diplôme d'Ingénieur) – Major in AI & Digital Health

- Selected coursework: Generative AI, Machine Learning, AI & Digital Health, Probabilistic Mathematics.

Centrale Digital Lab · Lille, France *Sep. 2025 – Aug. 2026*
Gap Year Program – Graduate Certificate in Artificial Intelligence

- Specialized curriculum combining 3 industrial engineering projects and a 6-month international internship.

Lycée Hoche · Versailles, France *Sep. 2021 – Sep. 2023*
*Classes Préparatoires aux Grandes Écoles (CPGE) – MP2I / PSI**

- Intensive 2-year undergraduate STEM preparation in Mathematics, Physics & Computer Science for competitive entrance exams to top French Grandes Écoles.

Technical Skills & Languages

- **Deep Learning & Computer Vision: PyTorch**, Transformers (**SwinUNETR**), 3D Computer Vision, Scikit-learn.
- **Generative AI & RAG:** RAG Architectures *from scratch*, **LLMs**, ChromaDB, Pinecone, BGE-M3, **Docling**.
- **Programming & Data: Python** (advanced), **SQL**, C, Java, JavaScript, PHP (basics), REST APIs, Webhooks.
- **Tools & Environments: Docker**, CI/CD, Linux (Bash), **Git / GitHub**, n8n.
- **Languages: English** (Fluent, TOEIC 910/990, 6-month immersion in Malaysia); **French** (Native); **German** (B1 level).

Extracurricular Activities & Leadership

- **Centrale Lille Student Union (Logistics Vice-Lead):** Managed large-scale events for 1,000+ students and oversaw an annual budget of 135 k€ (2024–2025).
- **Competitive Football (Soccer):** Regional-level goalkeeper for 17 years – leadership, resilience, and high-pressure communication.
- **Community Engagement (Tutoring):** Weekly volunteer academic tutoring in sciences and French for two Ukrainian refugees (since 2021).